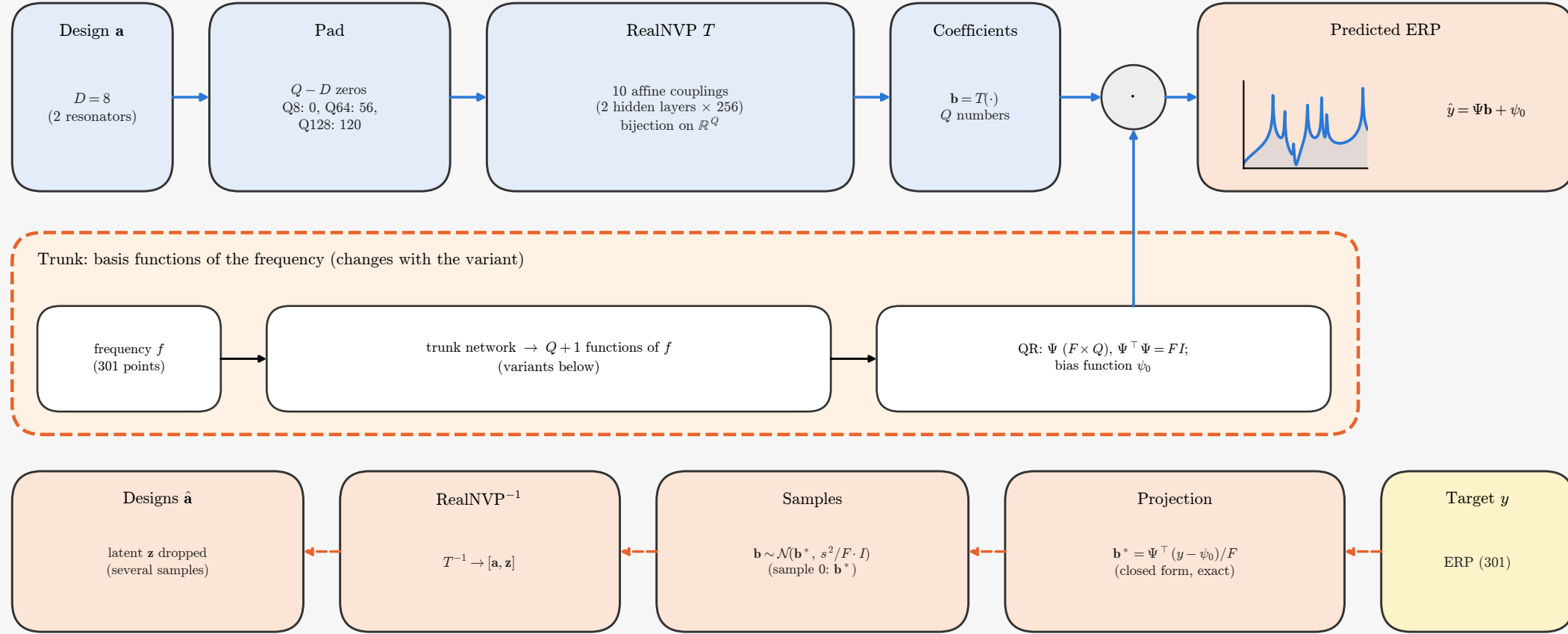


Invertible DeepONet: common structure

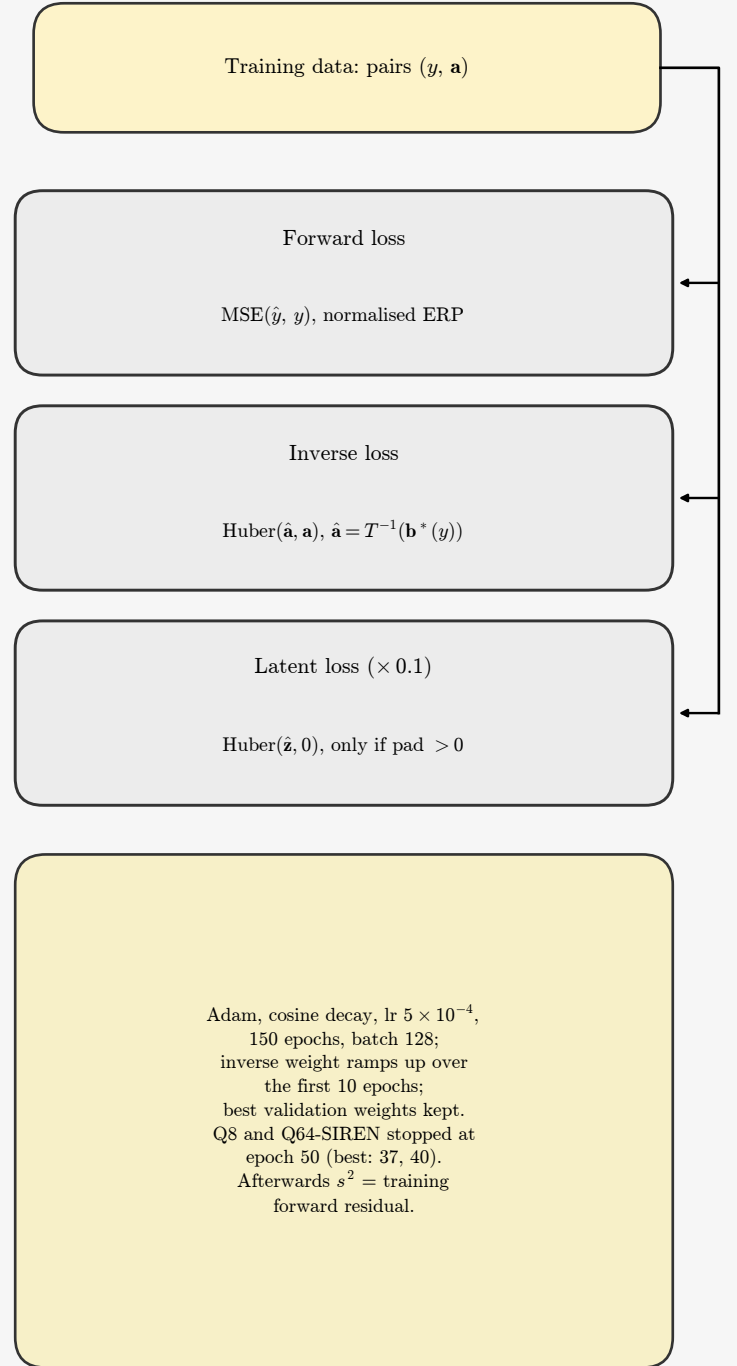
design $\leftrightarrow$ ERP through a bijection (RealNVP) and basis functions of f ; only the trunk changes between variants



Trunk variants: $Q = 64$ with the same RealNVP branch (1,153,280 parameters)

MLP trunk	FNO trunk	DCO trunk	DNO trunk	WNO trunk	LNO trunk	SIREN trunk
Fourier features (32 frequencies) $\rightarrow$ MLP, 4 layers $\times$ 256, SiLU	Fourier features $\rightarrow$ lift (40) $\rightarrow$ 4 Fourier blocks (32 modes, padded) $\rightarrow$ MLP	Fourier features $\rightarrow$ lift (128) $\rightarrow$ 4 residual MLP blocks $\rightarrow$ local refinement $\rightarrow$ linear	Fourier features $\rightarrow$ lift (96) $\rightarrow$ 4 FiLM blocks (FiLM from f) $\rightarrow$ refinement	Fourier features $\rightarrow$ lift (40) $\rightarrow$ 4 wavelet blocks (3-level Haar) $\rightarrow$ MLP	48 learned poles and residues (the same for every design) + Fourier features $\rightarrow$ lift (96) $\rightarrow$ MLP	$f \rightarrow$ 4 sine layers (128, $\omega_0 = 10$) $\rightarrow$ local refinement $\rightarrow$ linear
trunk 230,977 total 1,384,257	trunk 235,665 total 1,388,945	trunk 248,769 total 1,402,049	trunk 172,097 total 1,325,377	trunk 166,385 total 1,319,665	trunk 87,041 total 1,240,321	trunk 156,993 total 1,310,273

Loss functions



A trunk must not see the design, otherwise the inverse is no longer a projection. Size variants with the MLP trunk: Q8 (938,665 parameters), Q64 (1,384,257), Q128 (1,893,505).

Data: 200k configurations, 2 resonators (m, f_t, x, y by Latin hypercube, $k = m(2\pi f_t)^2$), 18 plate modes. Design in bounded logit coordinates, sorted by f_t .